

Ritvik Sharma

New York, NY

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Education

Columbia University

Aug 2025 - Dec 2026

MS in Computer Science - AIML & NLP Track

SRM University – Rank: 2/700+

Jun 2019 - May 2023

BTech in Computer Science - AI & ML Track

GPA: 9.86/10

Work Experience

Barclays, UK

Jul 2023 - Jul 2025

Software Engineer

Pune, India

- **Saved \$200K/year** by engineering AWS Lambda optimization utilities and **driving adoption** across Dev/DevOps.
- Achieved **96% faster performance (12 hrs → 27 mins)** for **5M+ record** ETL pipelines by redesigning Glue → DynamoDB workflows using Spark, Dynamic Frames, Batch Write; **Coordinated** with platform teams for seamless deployment.
- **Reduced error-resolution times by 75%** by designing and developing a complex error-routing utility via Python-based Lambdas, IBM Netcool, and SNS filters. Played a **key role in facilitating communication** between development and operations teams to align on system-wide logging and monitoring strategies.
- **Executed No-Downtime Deployments:** Integrated Blue-Green-Canary strategies into the system, **advised leadership, aligned cross-domain teams**, and ensured uninterrupted user experience.
- Improved CI/CD pipeline **success rates by 63%** by streamlining workflows, enhancing Dev ↔ DevOps collaboration.
- **Spearheaded the engineering and development** of the integration layer connecting Onboarding, Rewards, Payments, and other modules for Barclays' first cloud-based credit card platform, **ArrowCard** targeting **30M+ UK customers**.

Projects

VELVÉT - Luxury E-Commerce Website — AWS (SageMaker, OpenSearch), React, JavaScript, Python, NLP

[Link to Live Demo](#)

- Created full-scale real-world functionality, including **ML-based semantic product search, wishlist management, cart operations, dummy order placement, and similar product recommendations** for a seamless user experience.
- Implemented a secure and high quality AWS backend via IAM Roles, Basic DDoS Prevention and performed **Selenium-based unit + integration tests** to ensure Production-Grade Quality.

Columbia x Amazon Hackathon Winner: MyCFO.ai — AWS (Bedrock, Api-Gateway), TypeScript

[Link to Web/Video Demo](#)

- **Engineered a multi-agent CFO system** on Amazon Bedrock with a Supervisor Agent coordinating specialized financial agents, and **developed an interactive frontend** that transforms their structured outputs into real-time dashboards.
- **Architected a serverless pipeline** with API Gateway → Lambda → Bedrock Agents, enabling spam-filtered requests, scalable orchestration, and structured JSON outputs for lightweight frontend rendering.
- **Collaborated & Led a team of 4** in a 12-hour hackathon by strategically allocating tasks based on **analyzing individual strengths**, demonstrating exceptional **teamwork and impactful communication**.

Technical Skills

Languages: Python, JavaScript, Java, C/C++, SQL (Postgres, MySQL), NoSQL, HTML/CSS

Frameworks: React, TypeScript, Angular, Node.js, Flask, JUnit, JQuery, FastAPI, Tomcat, Django, Selenium, Flutter

Developer Tools: Git, GitHub, Jenkins, Docker, Amazon Web Services (AWS), GitLab, Linux, Kubernetes

Technologies: PyTorch, NLP, Pandas, NumPy, Scikit-Learn, Machine Learning, Computer Vision, OpenCV, matplotlib, LLM

Extracurriculars

- Sharma, Ritvik (April 2025). "VectorQuest: Faster and Better Open-Domain Question-Answering System"
Published in Springer Nature Switzerland AG for the innovation leading to **50 times Faster Prediction**.
- **2× Hackathon Winner** in university-level and industry-sponsored competitions, recognized for building innovative, production-grade AI and full-stack solutions under tight deadlines.
- **Providing one-on-one tutoring** to Student Athletes at Columbia in quantitative and computational subjects, including **Probability, Machine Learning, Calculus, and Data Structures & Algorithms**.